

Research Statement

My research focuses on developing machine learning approaches to analyze complex biological datasets and support translational biomedical discovery. During my Ph.D. in Computational Biology at the Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), under the supervision of Prof. Gajendra P. S. Raghava, I developed predictive computational models for understanding protein and peptide function in immunological and disease contexts. My work involved curating large experimental datasets, designing machine learning models, and deploying reproducible bioinformatics tools used by experimental researchers. This research resulted in multiple peer-reviewed publications and tools for predicting B-cell epitopes, identifying antigenic regions associated with Type-1 diabetes, and predicting antifreeze proteins and protein thermostability.

More recently, my work has focused on representation learning approaches for biological sequences using protein language models such as ProtBERT and ESM2. By integrating these embeddings with interpretable machine learning methods, I investigate how sequence representations capture functional and structural signals in proteins and how they can support predictive modeling and biological discovery. This work combines scalable computational pipelines, biological dataset integration, and modern deep learning approaches.

At Duke, I am particularly interested in extending these approaches toward artificial intelligence systems that support biomedical discovery and digital health applications. The focus on self-learning AI systems, autonomous experimentation frameworks, and multi-agent coordination for dynamic healthcare environments strongly aligns with my interest in building scalable AI systems that integrate heterogeneous biomedical data and support clinical decision-making. I aim to contribute to the development of reliable AI-driven tools that enable data-driven discovery and translational research in healthcare settings.